

ALEX YAN

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science

May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Computer Architecture, Programming in C, Discrete Mathematics, Linear Algebra, Introduction to Statistics

EXPERIENCE

Shopify

New York, NY

Software Engineer Intern

Sep. 2026 – Dec. 2026

- Incoming Fall Software Engineer Intern on the Shop Recognition & Continuity team.

PNC Bank

Pittsburgh, PA

Software Engineer Intern

Jun. 2026 – Aug. 2026

- Delivered Spring Boot validation features in Java for loan origination workflows on the application entry REST API, deploying through Red Hat OpenShift across testing environments.
- Expanded JUnit test coverage for validation logic, verifying valid, invalid, and edge-case scenarios end-to-end with Postman against configuration objects edited live in MongoDB Compass.
- Diagnosed and resolved intermittent Jenkins CI/CD failures caused by race conditions in asynchronous test execution.
- Built an LLM agent that catches convention and coverage gaps in unit tests before PR submission, grounded in team testing patterns and design discussion to cut PRs needed for approval by 55%.

Purdue Stack

West Lafayette, IN

Full Stack Developer

Sep. 2025 – May 2026

- Shipped a research collaboration platform in Next.js, FastAPI, and PostgreSQL for Purdue's Statistics Department, featuring role-based dashboards, real-time messaging, and JWT authentication. (Spring 2026)
- Cut TA scheduling setup time 25% with a preference-weighted greedy matching algorithm in Python, powering an assignment system for 150–200 TAs per semester in Purdue's Mathematics Department. (Fall 2025)

Rooh (Berkeley SkyDeck Pad-13 Incubator)

Berkeley, CA (Remote)

Software Engineer Intern

Jun. 2025 – Dec. 2025

- Cut voice response latency 20% by rearchitecting an end-to-end Google Cloud STT → LLM → TTS pipeline for a real-time conversational AI app in React and Next.js.
- Architected Next.js backend data pipelines supporting a live AI simulator and async review tool, handling real-time audio streaming, buffering, and transcription at scale.

PROJECTS

Perch | *Electron, TypeScript, React, SQLite, OpenAI API, WebSockets* | [Github](#)

Jul. 2026

- Built a local-first desktop focus app that generates a personalized AI mascot from a user's photo and delivers real-time, task-aware nudges on detecting off-task activity, tracking native window state and browser events through a paired Chrome extension.
- Designed a tiered classification pipeline escalating ambiguous states to OpenAI vision, with structured output validation, static fallbacks, and non-repeating message pools for reliable UX under API failure.

K/V Server | *Go, RPC, Concurrency*

Jul. 2026

- Built a linearizable key/value server and distributed lock in Go for MIT's distributed systems course (6.5840), using version-conditioned puts to guarantee at-most-once writes.

PyroTech | *Next.js, TypeScript, Google Gemini, WebSocket* | [Github](#)

Apr. 2026

- Developed an AI wildfire incident command simulation platform at UCLA's hackathon, with a Google Gemini orchestration layer coordinating 7 specialized agents across disaster, evacuation, resource, and infrastructure domains.
- Engineered a physics-based fire spread engine with dozer barrier and suppression zone modeling, streamed live over WebSocket atop real-world geospatial data for branching what-if scenario analysis.

Access Prep | *React, TypeScript, Flask, LangChain, Gemini* | [Github](#)

Jan. 2026

- Won **3rd Place** at the Capital One Tech Summit Hackathon by building a bilingual document preparation platform serving underrepresented populations across 5 immigration and financial aid pathways.
- Implemented a Flask REST API handling session management, file uploads, and AI validation, integrating LangChain and Gemini to power a context-aware chatbot guiding users through the platform.

SKILLS

Languages: Python, Go, Java, JavaScript, TypeScript, C, C++, SQL, Bash

Frameworks: React, Next.js, Node.js, Flask, FastAPI, Spring Boot, Tailwind CSS

Tools & Testing: Git, Bitbucket, GitHub Actions, Jenkins, Gradle, JUnit, Postman

Databases & Cloud: PostgreSQL, MongoDB, AWS (Lambda, API Gateway), Google Cloud, Docker, Kubernetes/OpenShift

AI Dev Tools: Claude Code, GitHub Copilot, Cursor, Codex, Antigravity